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Roll No

EX-6001-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Communication Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Find the Fourier Transform of the following functions and draw the spectrum: 7
i) $g(t) = e^{-bt} u(t)$
ii) Constant function A
b) Give classification of signals. How can you differentiate energy and power signals? 7
2. a) Explain the following terms: 7
i) Auto Correlation
ii) Standard Deviation
iii) Variance
b) Explain companding. Also describe A-law and μ -law. 7
3. a) Define Modulation Index. Explain pre-emphasis and de-emphasis in FM. 7
b) With the help of circuit diagram, explain the working of balanced modulator for DSB-SC generation. 7

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4. a) Explain the demodulation of AM wave by envelope detector. Draw the circuit and necessary waveforms. 7
b) What do you understand by angle modulation? Explain the types of angle modulation and relationship between them. 7
5. a) Explain the working of superheterodyne receiver. 7
b) With the help of block diagram, explain AFC system. 7
6. a) What are the processes involved in PCM? What do you mean by “Quantization”? State the advantages of PCM. 7
b) Draw a schematic diagram of QPSK generating system and explain its working. 7
7. a) Explain the advantages of TDMA over FDMA. 7
b) Define the following terms: 7
 - i) EIRP
 - ii) Effective aperture and aperture efficiency
8. Write short notes on (any two): 14
 - a) IF Amplifiers
 - b) Earth station
 - c) Transponders
 - d) Convolution and its properties

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